



GIRIJANANDA CHOWDHURY UNIVERSITY, ASSAM

Hatkhowapara, Azara, Guwahati 781017, Assam

BMA23113T	Mathematics to Civil Engineering	L	T	P	C
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Pre-requisite: Physics, Mathematics I and II					
Course Objectives					
- To enable the students to study Fourier Transforms of various functions - To enable the students to study the Laplace Transforms, properties of Laplace Transform, inverse Laplace Transform and some applications to solve the differential equations and integral equations - To find the solutions of second order PDEs by the technique of separation of variables determined by conditions at the boundary of the spatial domain and initial conditions at time zero.					
Course Outcome					
After successful completion of this course, the students will be able to					
CO1: apply basic knowledge of Fourier series and develop Fourier series of periodic functions					
CO2: evaluate Laplace Transform as well as Inverse Laplace Transform of function and solve the ordinary differential equations and their applications to engineering sciences.					
CO3: apply the separation of variables method to solve well-posed initial and boundary value problems for PDEs					
Module 1: Fourier Series					14 hours
Definition of Fourier series, Orthogonal and Orthonormal functions, Fourier series with arbitrary period, in particular periodic function with period 2, Fourier series of even and odd function, Half range Fourier series.					
Module 2: Laplace Transforms and Applications					20 hours
Introduction, Definition of the Laplace transform, Useful properties of Laplace transform (without proof): Linearity, First shifting theorem, Multiplication and division by t, transforms of derivatives and integrals, Heaviside unit step function, Dirac’s delta function, second shifting theorem, Laplace transform of Periodic function, Inverse Laplace transform using partial fraction and Convolution theorem (without proof), Application to solve initial and boundary value problem involving ordinary differential equations with one dependent and constant coefficient.					
Module 3: Partial Differential Equation					14 hours
Second order PDE of mathematical physics (Heat, wave and Laplace equation, one dimensional with standard boundary conditions), Solution by separation of variable method using Fourier series.					
Total Lecture hours					48 hours



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Text Book(s)

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| 1 | AICTE's Prescribed Textbook: Mathematics-I (Calculus & Linear Algebra), Reena Garg, Khanna Book Publishing Co., New Delhi, 2023. |
| 2 | Bali, N. P., Goyal M., A text book of engineering Mathematics, Laxmi Publications, Reprint, 2014 |

Reference Book(s)

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| 1 | Kreyszig E, Advanced Engineering Mathematics, 9th Edition, John Wiley & Sons, 2006. |
| 2 | Garg R., Engineering Mathematics, Khanna Book Publishing Company, 2022. |
| 3 | Wiley C. R., "Advanced Engineering Mathematics", McGraw Hill Inc., New York Ed., 1993 |
| 4 | Ramana D. V., "Higher Engg. Mathematics", The McGraw-Hill Inc., New Delhi, 2007. (ISBN 007063419X) |